Activity Proposal for IT Olympics (Cybersecurity) Training Purposes

Activity Description

- Hands-On Practice: Laptop-to-Switch Connection, Kali Linux VM Configuration, and Basic Network Troubleshooting
- This activity gives participants practical experience building a small working network and preparing a Kali Linux virtual machine for cybersecurity exercises. Participants work in groups and complete the following tasks:
- Connect a laptop to a network switch using an Ethernet cable.
- Configure the Kali Linux VM inside the laptop to connect to the network, including virtualization software setup, bridged network adapter, and IP addressing.
- Configure everything else needed for the network to work, including the switch, IP scheme, and host and VM settings, and verify connectivity with other devices.
- Perform basic network troubleshooting, including checking cables and link lights, verifying IP settings, testing with ping and traceroute, and resolving common faults.
- By the end of the activity, each group should have a laptop and Kali Linux VM that can communicate with other devices on the network and can diagnose and fix common connectivity problems.

Equipment Needed

One (1) Laptops with virtualization support enabled (VT-x/AMD-V), at least 8 GB RAM, 4 cores, and 50 GB free storage
Two (2) Network Switches, 24-port, managed or unmanaged
Two (2) Ethernet (UTP) Cables, Cat5e or Cat6, 15 meters each
One (1) Router or Gateway, if Internet or DHCP access is required
One (1) Console/Rollover Cable, with one (1) USB-to-Serial Adapter, if configuring a managed switch
One (1) Extension Cord or Power Strip
One (1) Cable Tester, for troubleshooting practice

Facilities Needed

One (1) Computer Laboratory or Training Room
One (1) Table or Workstations, one for each group's laptop and switch connection
Reliable electrical supply with enough outlets for all laptops, switches, and extension cords
Wi-Fi or Internet access for downloading updates and tools, if needed

Other Material/Software/Infrastructure Needed

Virtualization Software: VirtualBox, VMware Workstation Player, or Hyper-V
Kali Linux: One (1) pre-built VM image or ISO, downloaded beforehand and copied to each laptop
One (1) IP Addressing Plan and Network Diagram, with assigned addresses per group (e.g., 192.168.1.0/24)
One (1) Switch Configuration Guide, with vendor documentation and admin credentials if the switch is managed (If Available)
Troubleshooting Tools: ping, traceroute, ipconfig/ifconfig/ip, nslookup, Wireshark, and nmap
One (1) Printed or Digital Lab Guides with step-by-step instructions and a troubleshooting checklist (Tentative)
One (1) Set of Pre-Injected Faults (e.g., wrong IP, wrong adapter mode, unplugged cable) for the troubleshooting portion
One (1) Closed Lab Network, or firewall rules, so the practice activity does not affect production systems
Two (2) USB Flash Drives (32 GB or larger) containing ISOs, VM images, and tools for offline installation

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